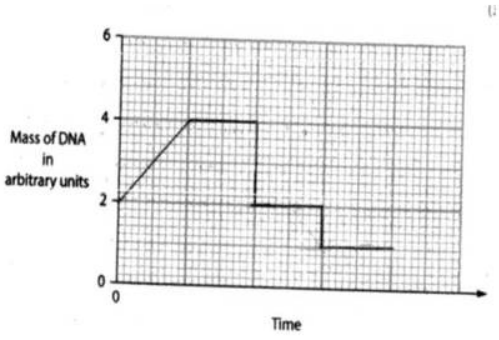
Type MS · Locate DNA and chromosomes in a human cell

Answer 1 of 6 · Question 4 part a.i · 1 mark

2023 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number				
4	(a) (i)	**A; (chromosomes)** B/C/D do not contain DNA		1 1
	(ii)	RNA/mRNA/tRNA;		
	(iii)	<table><tr><th>Differences</th><th>DNA</th><th>Named nucleic acid</th></tr><tr><td>1</td><td>double stranded</td><td>single stranded</td></tr><tr><td>2</td><td>contains deoxyribose</td><td>contains ribose</td></tr><tr><td>3</td><td>contains thymine</td><td>contains uracil</td></tr></table>		3
	(b) (i)	• replication; • formation of a new strand (of DNA); • by complementary base pairing; • double the original mass of DNA;		4
	(ii)	• cell dividing; • cytokinesis;		2
	(iii)	• line starting at two; • increases to four; • ends at one;  		3

Total 14 marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 1 of 6 · Question 4 part a.i · 1 mark

2023 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 2 of 6 · Question 5 part a.i · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
5	(a) (i)	A (DNA); B/C/ and D are found in or associated with other structures.		1
	(ii)	correct circles of pair;		1
	(iii)	only two pairs of homologous chromosomes/ would be 23 in a human cell;		1
	(b)	• prophase, metaphase, anaphase and telophase; • (prophase) chromosomes become thicker/visible; • (metaphase) chromatids line up at equator; • (anaphase) chromatids separate and migrate to poles; • (telophase) nucleus/nuclear membrane reforms;	Allow condense Allow chromosomes throughout Allow clear annotated diagrams showing stages	5

Total 8 marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 2 of 6 · Question 5 part a.i · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 3 of 6 · Question 7 part b.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
7 (a)	(i)	• prophase, metaphase, anaphase and telophase;;	minus 1 for each error	2 marks
	(ii)	• chromosomes lined up; • equator shown/aligned at equator; • one member of each bivalent above, the other below;		3 marks
(b)	(i)	A (DNA); B not part of DNA C not part of DNA D found only in RNA not DNA		1 mark
	(ii)	length = 20 mm; $\frac{20}{750}$ = 0.026 mm; = 26.0/26.7 µm;	allow 20 -22 ECF	3 marks
(c)		• two genetically identical cells; • diploid/2n;		2 marks

Total 11 marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 3 of 6 · Question 7 part b.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 4 of 6 · Question 1 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Locate DNA and chromosomes in a human cell

Answer 4 of 6 · Question 1 part a.iv · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 5 of 6 · Question 3 part a.i · 1 mark

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

3 (a)	(i)	nucleus;		(1)
	(ii)	allele;		(1)
	(iii)	too many fingers/toes;		(1)
(b)	(i)	heterozygous/heterozygote/carrier;		(1)
	(ii)	Y;		(1)
(c)		Any four from: • double helix/two strands; • consists of nucleotides; • four bases; • each strand held together by hydrogen bonds/bases linked by hydrogen bonds; • complementary base pairing/A-T, G-C;		(4)

Total Question 3 = 9 marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 6 of 6 · Question 2 part a.iv · 1 mark

2025 JUN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

2 (a)	(i)	X = (pentose) sugar/deoxyribose; Y = phosphate;	reject ribose reject phosphorus	2
	(ii)	DNA/deoxyribonucleic acid;		1
	(iii)	• double/two stranded; • thymine present; • only found in DNA (not RNA);		3
	(iv)	nucleus;		1
(b)	(i)	hydrogen bonds;		1
	(ii)	• connect/hold/join, two strands/bases, together; • between complementary bases (pairs)/nucleotides; • adenine thymine/guanine cytosine;	ignore just A T G C	3
				11 marks

Type MS · Locate DNA and chromosomes in a human cell

Answer 6 of 6 · Question 2 part a.iv · 1 mark

2025 JUN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks